Geometrically

How to compute?
To find:

$$\frac{\partial f}{\partial x} = f_x$$

Treat y as a function, x as variable.

Example

$$f(x, y) = x^3y + y^2$$

$$\frac{\partial f}{\partial x} = 3x^2y$$

$$\frac{\partial f}{\partial y} = x^3 + 2y$$

Partial Derivatives

For $f(x, y)$:

$$\frac{\partial f}{\partial x} = f_x \quad \text{(vary } x\text{, keep } y \text{ constant)}$$

$$\frac{\partial f}{\partial y} = f_y \quad \text{(vary } y\text{, keep } x \text{ constant)}$$

Approximation Formula

If we change:

$$x \to x + \Delta x, \quad y \to y + \Delta y$$

Approximation and Tangent Plane

Let

$$z = f(x, y)$$

then

$$\Delta z \approx f_x \Delta x + f_y \Delta y$$

Justification (Tangent Plane)

We know that f_x and f_y are slopes of two tangent lines.

At the point (x_0, y_0):

$$\frac{\partial f}{\partial x}(x_0, y_0) = a$$

gives tangent line:

$$L_1 : \begin{cases} z = z_0 + a(x - x_0) \\ y = y_0 \end{cases}$$

$$\frac{\partial f}{\partial y}(x_0, y_0) = b$$

gives tangent line:

$$L_2 : \begin{cases} z = z_0 + b(y - y_0) \\ x = x_0 \end{cases}$$

Both L_1 and L_2 are tangent to the surface

$$z = f(x, y)$$

Together, they determine the tangent plane:

$$z = z_0 + a(x - x_0) + b(y - y_0)$$

Interpretation

The approximation formula says that the graph of f is close to its tangent plane near (x_0, y_0).

Application of Partial Derivatives

Optimization Problems

Find minimum or maximum of a function $f(x, y)$.

Optimization by Completing the Square

$$f(x, y) = (x - y)^2 + 2y^2 + 2x - 2y$$

Complete the square:

$$f(x, y) = \big[(x - y) + 1\big]^2 + 2y^2 - 1$$

$$\big[(x - y) + 1\big]^2 \geq 0, \quad 2y^2 \geq 0$$

$$\Rightarrow f(x,y) \geq -1$$

So the minimum value is:

$$f_{\min} = -1$$

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Least Squares Interpolation

Given experimental data:

$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$$

Find the best fit line:

$$y = ax + b$$

We determine a and b by minimizing the total squared error.
For each data point:

$$\text{error} = y_i - (ax_i + b)$$

Define:

$$D = \sum_{i=1}^{n} \left[y_i - (ax_i + b)\right]^2$$

Minimize D:

$$\frac{\partial D}{\partial a} = \sum_{i=1}^{n} 2\left(y_i - (ax_i + b)\right)(-x_i) = 0$$

$$\frac{\partial D}{\partial b} = \sum_{i=1}^{n} 2\left(y_i - (ax_i + b)\right)(-1) = 0$$

Mathematics Notes

Complete the Square

$$f(x,y) = (x-y)^2 + 2y^2 + 2x - 2y$$

$$f(x,y) = \big[(x-y)+1\big]^2 + 2y^2 - 1$$

$$f(x,y) \geq -1$$

Minimum value: -1

Least Squares Interpolation

Given experimental data:

$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$$

Find the best fit line:

$$y = ax + b$$

We minimize the total squared error:

$$Q = \sum_{i=1}^{n} [y_i - (ax_i + b)]^2$$

Taking partial derivatives:

$$\frac{\partial Q}{\partial a} = \sum_{i=1}^{n} 2\,(y_i - (ax_i + b))\,(-x_i) = 0$$

$$\frac{\partial Q}{\partial b} = \sum_{i=1}^{n} 2\,(y_i - (ax_i + b))\,(-1) = 0$$

Normal Equations

From minimizing Q, we obtain:

$$\sum_{i=1}^{n} x_i\big(ax_i + b - y_i\big) = 0$$

$$\sum_{i=1}^{n} \big(ax_i + b - y_i\big) = 0$$

These give the system:

$$\left(\sum_{i=1}^{n} x_i^2\right) a + \left(\sum_{i=1}^{n} x_i\right) b = \sum_{i=1}^{n} x_i y_i$$

$$\left(\sum_{i=1}^{n} x_i\right) a + nb = \sum_{i=1}^{n} y_i$$

This is a 2×2 linear system. Solve for a and b.

It can be shown this gives a minimum.

Extensions of Least Squares

1. Exponential Fit

$$y = ce^{ax}$$

Take logarithm:

$$\ln(y) = \ln(c) + ax$$

This reduces to a linear form, so we can apply least squares.

2. Quadratic Fit

$$y = ax^2 + bx + c$$

Minimize:

$$Q(a, b, c) = \sum_{i=1}^{n} \big[y_i - (ax_i^2 + bx_i + c)\big]^2$$

Critical Points

Critical points of $f(x, y)$ occur where:

$$f_x = 0 \quad \text{and} \quad f_y = 0$$

Questions:

- How do we decide between local minimum, local maximum, or saddle point?
- How do we find global minimum/maximum?

Global extrema occur either at a critical point or on the boundary (or at infinity).

Second Derivative Test

First consider:

$$w = ax^2 + bxy + cy^2$$

Example:

$$w = x^2 + 2xy + 3y^2$$

$$= (x + y)^2 + 2y^2$$

General case: If $a \neq 0$

$$w = a\left(x^2 + \frac{b}{a}xy\right) + cy^2$$

Complete the square:

$$w = a\left(x + \frac{b}{2a}y\right)^2 + \left(c - \frac{b^2}{4a}\right)y^2$$

Equivalent form:

$$w = \frac{1}{4a}\left[(2ax + by)^2 + (4ac - b^2)y^2\right]$$

3 Cases

1. $4ac - b^2 < 0$
   One term ≥ 0, the other ≤ 0
2. $4ac - b^2 = 0$
   e.g. $w = x^2$
   Degenerate critical point

3. $4ac - b^2 > 0$

$$w = \frac{1}{4a}\left[(\cdot)^2 + (\cdot)^2\right]$$

If $a > 0$, minimum
If $a < 0$, maximum

$$w = a\left(x + \frac{b}{2a}\right)^2$$

Quadratic formula:

$$b^2 - 4ac$$

$$w = ax^2 + bxy + cy^2$$

$$w = y^2\left[a\left(\frac{x}{y}\right)^2 + b\left(\frac{x}{y}\right) + c\right]$$

If $b^2 - 4ac \geq 0$, then this takes both positive and negative values.
$\Rightarrow w$ takes positive and negative values $\Rightarrow$ saddle

If $b^2 - 4ac < 0$

$$a\left(\frac{x}{y}\right)^2 + b\left(\frac{x}{y}\right) + c$$

or always $> 0 \Rightarrow w \geq 0$
always $< 0 \Rightarrow w \leq 0$

In general, look at second derivatives:

$$\frac{\partial^2 f}{\partial x^2} = f_{xx}$$

$$f_{xy} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = f_{yx}$$

$$\frac{\partial^2 f}{\partial y^2} = f_{yy}$$

Second derivative test
At a critical point (x_0, y_0) of f:
Let

$$A = f_{xx}(x_0, y_0), \quad B = f_{xy}(x_0, y_0), \quad C = f_{yy}(x_0, y_0)$$

If

$$AC - B^2 > 0 \quad \text{and} \quad A > 0$$

then there is a local minimum.

$$AC - B^2 > 0 \text{ and } A > 0 \quad \Rightarrow \text{local minimum}$$

$$AC - B^2 < 0 \quad \Rightarrow \text{saddle point}$$

$$AC - B^2 = 0 \quad \Rightarrow \text{cannot conclude}$$

Verify in a special case:

$$w = ax^2 + bxy + cy^2$$

$$w_x = 2ax + by, \quad w_y = bx + 2cy$$

$$w_{xx} = 2a, \quad w_{xy} = b, \quad w_{yy} = 2c$$

$$A = 2a, \quad B = b, \quad C = 2c$$

$$AC - B^2 = 4ac - b^2$$

Quadratic approximation

$$\Delta f \approx f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

$$+\frac{1}{2} f_{xx}(x - x_0)^2 + f_{xy}(x - x_0)(y - y_0) + \frac{1}{2} f_{yy}(y - y_0)^2$$

So the general case reduces to the quadratic case.